

KUMAGAYA, Japan

WMO: 476260

Lat: 36.1503N

Lon: 139.3806E

Elev: 32

StdP: 100.94

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.1	-1.2	-13.8	1.1	6.0	-12.2	1.3	5.3	9.2	7.1	8.5	7.0	2.0	290	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.4	35.7	25.5	34.2	25.2	32.7	24.6	26.6	32.8	26.0	31.8	25.5	30.7	3.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.3	28.6	24.6	19.7	28.4	24.1	19.0	28.3	83.2	32.9	80.9	31.9	78.6	30.8	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.0	6.8	5.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-4.2	38.0	0.7	1.1	-4.7	38.8	-5.1	39.5	-5.5	40.1	-6.0	40.9		
			-6.0	27.5	0.7	0.6	-6.5	27.9	-6.9	28.2	-7.3	28.6	-7.8	29.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	4.6	5.4	8.8	14.3	19.3	22.6	26.7	27.9	23.8	18.0	11.9	6.9
	DBStd	8.54	2.07	2.50	3.18	3.43	2.86	2.79	3.21	2.81	3.26	2.93	2.88	2.55
	HDD10.0	483	168	132	63	6	0	0	0	0	0	0	13	102
	HDD18.3	1827	426	363	296	127	23	2	0	0	2	41	193	355
	CDD10.0	2641	1	2	26	134	288	378	519	554	414	249	70	6
	CDD18.3	943	0	0	0	5	53	131	261	296	166	31	0	0
	CDH23.3	8337	0	0	2	52	373	848	2561	3169	1204	126	1	0
	CDH26.7	3342	0	0	0	6	83	256	1138	1431	409	21	0	0
(14) Wind	WSAvg	2.5	3.0	3.1	3.0	2.7	2.4	2.2	2.1	2.1	2.0	2.0	2.2	2.6
(15) Precipitation	PrecAvg	1375	40	39	71	95	121	153	176	190	218	166	67	39
	PrecMax	1877	109	123	130	224	240	385	333	433	419	518	270	155
	PrecMin	1012	3	5	26	23	38	51	72	30	28	18	2	4
	PrecStd	198	32	27	29	55	52	66	76	114	97	121	52	32
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	18.0	22.5	27.3	31.3	33.6	37.2	37.3	34.5	29.1	22.2	17.5
		MCWB	7.5	10.4	12.0	15.9	18.3	22.9	25.6	25.8	24.6	20.4	14.3	10.7
	2%	DB	12.3	14.3	19.3	24.7	28.7	31.2	35.7	35.9	32.7	26.4	20.3	15.0
		MCWB	5.5	7.1	10.6	14.4	17.6	21.5	25.5	25.6	23.9	18.8	13.5	8.7
	5%	DB	10.7	12.3	17.1	22.7	27.1	29.6	34.2	34.6	30.8	24.5	18.6	13.4
		MCWB	4.5	5.9	9.3	13.8	17.1	20.9	25.2	25.5	23.5	17.9	12.4	7.5
	10%	DB	9.3	10.5	14.9	20.8	25.4	27.9	32.4	33.2	29.0	22.9	17.1	11.8
		MCWB	3.7	4.7	8.6	13.1	16.8	20.4	24.6	25.2	22.4	16.9	11.9	6.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	11.5	14.1	19.5	21.7	25.1	27.2	27.4	26.1	22.2	16.8	12.2
		MCDB	11.6	16.7	18.5	22.8	26.1	30.2	34.1	34.1	31.1	26.0	19.6	14.9
	2%	WB	6.7	8.4	12.3	17.0	20.0	23.8	26.5	26.8	25.2	20.8	15.4	10.0
		MCDB	11.1	12.8	16.8	21.8	25.1	28.7	32.9	33.0	29.5	24.2	18.1	13.3
	5%	WB	5.2	6.8	10.8	15.6	19.0	22.8	25.9	26.2	24.6	19.6	14.1	8.6
		MCDB	9.6	10.8	15.5	20.3	23.8	27.1	31.9	31.9	28.6	22.8	16.6	12.2
	10%	WB	4.1	5.4	9.2	14.4	18.1	21.8	25.3	25.8	23.9	18.4	12.9	7.3
		MCDB	8.5	9.6	14.0	18.6	22.6	25.5	30.6	31.0	27.6	21.6	15.8	10.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.6	10.0	10.5	9.9	8.1	8.1	8.4	7.6	7.8	8.9	9.4
		MCDBR	11.4	12.5	13.4	13.7	13.2	11.1	10.9	10.7	10.0	9.9	10.6	11.1
	5% WB	MCWBR	6.7	7.3	7.4	6.6	5.4	4.1	3.4	3.1	3.4	4.5	5.7	6.5
		MCWBR	10.7	11.3	11.7	11.1	10.4	8.7	10.0	9.6	8.6	7.9	8.3	9.9
(40) Clear-Sky Solar Irradiance	taub	taub	0.365	0.411	0.481	0.538	0.570	0.616	0.608	0.605	0.525	0.477	0.410	0.368
		taud	2.242	2.105	1.948	1.868	1.847	1.802	1.873	1.869	2.042	2.099	2.214	2.278
	Ebn at Noon	Ebn at Noon	808	807	781	760	742	706	709	699	736	731	749	778
		Edh at Noon	106	136	174	199	206	216	200	197	157	136	109	96
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.67	3.30	4.13	4.80	5.16	4.59	4.71	4.70	3.75	3.08	2.54	2.35
	RadStd	RadStd	0.16	0.25	0.24	0.43	0.53	0.38	0.75	0.48	0.32	0.31	0.22	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page